

Data Visualization & Storytelling — Project Guide

This project focuses on creating clear, insightful, and compelling data visualizations using **Tableau** or **Power BI**. The goal is to transform raw data into meaningful business insights through effective visual storytelling.

Objective

To analyze the *Superstore.csv* dataset (or any similar sales dataset) and develop a visual report/dashboard that highlights key business insights through charts, trends, and contextual storytelling.

Tools Used

- **Power BI** (or Tableau)
- Dataset: **Superstore.csv**
- Deliverable: PDF report or dashboard screenshots

Project Workflow / Process

1 Load the Data

- Import the dataset into Power BI/Tableau.
- Verify data types for fields such as dates, numbers, and categories.

2 Data Exploration (EDA)

- Review key metrics: Sales, Profit, Quantity, Category, Region, Segment.
- Identify trends, patterns, or anomalies.
- Understand what questions the data can answer.

3 Choose the Right Visuals

Based on the analysis goals:

- Line chart → Sales trends over time
- Bar chart → Category or sub-category comparisons
- Map → Region-wise performance
- Scatter plot → Sales vs profit relationship

- Treemap → Contribution of categories

4 Build Visualizations

- Create clear and minimal charts.
- Avoid clutter and excessive colors.
- Add proper titles, labels, legends, and highlights.

5 Add Insights & Context

For each chart:

- Write brief explanations.
- Highlight trends (e.g., “Technology drives majority of profits”).
- Add reference lines or annotations where necessary.

6 Create the Dashboard

- Combine the key visuals into a single Power BI/Tableau dashboard.
- Ensure a clean layout and logical flow.
- Add filters for user interactivity (Region, Category, Year).

7 Summary / Storyboard

- Prepare a summary slide that captures:
 - Business problems
 - Findings
 - Actionable recommendations

8 Export Final Deliverable

- Export report as PDF or include dashboard screenshots.
- Upload to GitHub along with this README.

Key Insights (Example)

This section can be updated based on your actual findings.

- Sales increased steadily from 2012 to 2014.
- Technology category contributed the highest profit.
- West region performed the best overall.
- Some sub-categories (e.g., Tables) show negative profit trends.

Files Included

- `dashboard.pdf / report.pdf`

- PowerBI.pbix / Tableau.twbx
- README.md (this file)

Outcome

By completing this project, we practiced:

- Selecting appropriate charts
- Applying visual storytelling
- Converting data into insights
- Creating professional dashboards